

PHY1004W Problem set 07

a) Since the container is moving at a constant speed in the same direction, $\Rightarrow \vec{v}$ is constant.

$$\therefore \vec{F}_{\text{net}} = 0$$

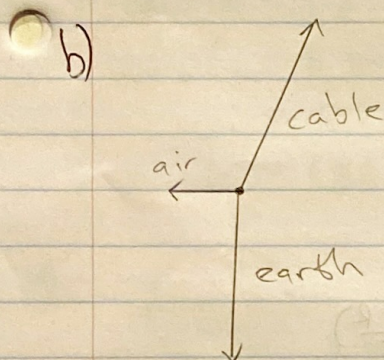
$$\therefore \frac{d\vec{p}}{dt} = \vec{F}_{\text{net}} = 0$$

Objects exerting a force:

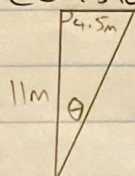
Earth (exerting gravitational force)

The cable (tension)

The air (exerting air friction)



c) Consider:



$$\theta = \arctan\left(\frac{4.5}{11}\right)$$

$$= 22.25^\circ$$

Hence since $\vec{F}_{\text{net}} = 0 \Rightarrow F_{\text{net},y} = 0$

$$\Rightarrow T \cos 22.25 = 400 \cdot 9.8 \quad (T_{\text{component}} = \text{weight})$$

$$\therefore T = \cancel{4522} 9530 \text{ N}$$

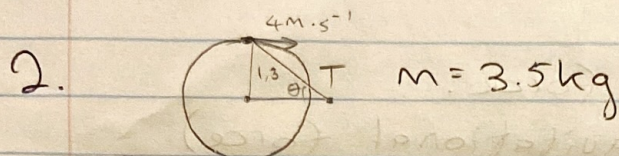
d) $\vec{F}_{\text{cable}} = \langle 9530 \sin 22.25, 9530 \cos 22.25, 0 \rangle \text{ N}$
 (with up as the positive $\hat{j}$ direction)
 $= \langle 3609, 8820, 0 \rangle \text{ N}$

e) $\frac{d\vec{p}}{dt} = 0 \Rightarrow \vec{F}_{\text{net}} = 0 \Rightarrow \vec{F}_{\text{net},x} = 0$

$$\therefore F_{\text{cable},x} = F_{\text{air}} \Rightarrow F_{\text{air}} = 3609 \text{ N}$$

f) $\vec{F}_{air} = \langle -3609, 0, 0 \rangle \text{ N}$

g) Yes, the breaking tension is 9300N whereas the tension in the cable is 9530 > 9300



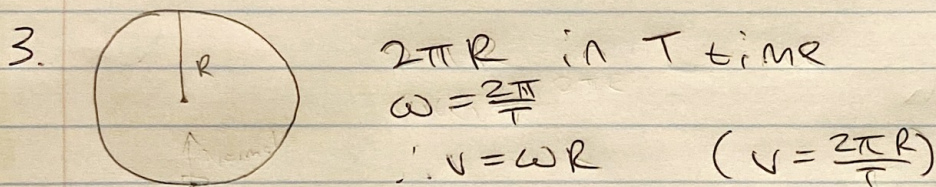
$F_{net, \perp} = T \sin \theta + mg = \frac{mv^2}{R}$ where θ is the angle between the rope and the $\hat{r}$ axis

Since $\frac{dp_z}{dt} = 0 \Rightarrow F_{net, z} = 0$

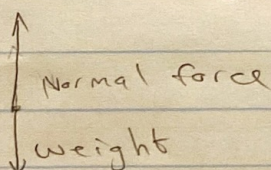
$\Rightarrow T \cos \theta = 0 \Rightarrow \cos \theta = 0 \Rightarrow \theta = 90^\circ$

$\therefore T \sin 90 + 3.5 \times 9.8 = \frac{3.5(4)^2}{1.3}$

$\therefore T = 8.78 \text{ N}$



Force diagram:



$F_{net, \perp} = N - mg = \frac{mv^2}{R} = \frac{m(\omega R)^2}{R} = m\omega^2 R$

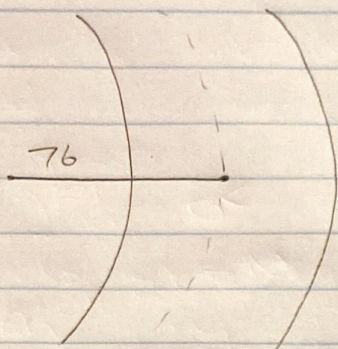
$\Rightarrow N = m\omega^2 R + mg$ and $\hat{N} = \langle 0, 1, 0 \rangle$

where up is the positive $\hat{j}$ direction

ω can be replaced with $\frac{2\pi}{T}$ giving:

$N = m\left(\frac{2\pi}{T}\right)^2 R + mg$
 $= \frac{4\pi^2 m R}{T^2} + mg$

4.



$$F_{\text{net}, L} = F_s$$

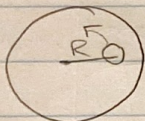
$$\Rightarrow \frac{mv^2}{R} = M_s g$$

$$v_{\text{max}}^2 = R M_s g$$

$$= 76 \times 0.25 \times 9.8$$

$$= 186.2 \text{ m} \cdot \text{s}^{-1} \Rightarrow v_{\text{max}} = 13.65 \text{ m} \cdot \text{s}^{-1}$$

5.



$$\omega = \frac{2\pi}{T} = 2\pi \cdot \frac{\text{rpm}}{60} = 2\pi \cdot \frac{33.3}{60} \quad \left(\frac{\text{rpm}}{60} = \frac{60}{T} \right)$$

$$= 3.49 \text{ rad} \cdot \text{s}^{-1}$$

Now since $v = \omega R$ ($v = \frac{2\pi R}{T}$)

$$\Rightarrow F_{\text{net}, L} = m\omega^2 R$$

$$\Rightarrow M_s g = m\omega^2 R$$

$$R = \frac{M_s g}{\omega^2}$$

$$= \frac{0.1 \times 9.8}{3.49^2} = 0.0805 \text{ m}$$